ARCHIT SHARMA

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EDUCATION

New York University — Sept 2024 - May 2026
M.S. Mechatronics and Robotics | Mobile Robotics — *GPA: 3.8*
Birla Institute of Technology Pilani - Goa — Aug 2019 - May 2023
B.Tech Mechanical Engineering — *GPA: 3.4*

TECHNICAL SKILLS

Programming & Software Python, C/C++, Linux, Git, Docker, MATLAB/Simulink, Cloud & HPC (AWS, GCP, SLURM), CI/CD & Testing

ML & Perception PyTorch, SLAM, Imitation Learning, Behavior Cloning, Diffusion Policy, VLA, PPO, State Estimation (EKF/UKF, Kalman Filter), Sensor Fusion, OpenCV, RL Frameworks (SB3, RLlib), Edge Deployment, YOLO, ResNet, 3D Reconstruction (NeRF, Gaussian splatting)

Robotics, Control & Simulation ROS/ROS2, NVIDIA Isaac Sim/Isaac Lab, MuJoCo, Sim-to-Real Transfer, Motion Planning, Optimal Control (MPC, MPPI), Inverse Kinematics (Mink, Pink), Classical Control (PID, LQR), Point-Cloud Registration (ICP/RANSAC), Convex Optimization (OSQP), Photogrammetry, Unity, Unreal Engine, Multiphysics Simulation (COMSOL, ANSYS)

Embedded Systems & Hardware NVIDIA Jetson, Raspberry Pi, Arduino, ESP32, Firmware Development, Real-Time Control (RTOS), I2C/SPI/CAN, CAD(SolidWorks, OnShape), CNC Machining

PUBLICATIONS

VICTR: Value-Informed Context Retrieval for Vision-Language-Action Models
L. Macesanu, J. Zhang, T. Zhang, A. Patil, K. Angers, **A. Sharma**, K. Imam, G. Perez-Haiek, C. Clark, C. Feng
Submission targeting ICRA 2027

Neural Network-Based Control for Vehicle Suspension Systems with Active Damping
Archit Sharma, Pravin M Singru
The 30th International Congress on Sound and Vibration, 2024

EXPERIENCE

Graduate Researcher, AI4CE Lab — July 2026 - Present
- Designed an auto-annotation pipeline to train an in-context-learning value function estimator for a VLA, building a generative reward model based on robo-dopamine, achieving ∼29% higher correlation with manual annotations than robometer.
- Built a 3M-frame teleoperated dataset spanning more diverse manipulation tasks than benchmarks like LIBERO, teleoperating a bimanual YOR robot.
- Benchmarked pi0-FAST and pi0.5 on the robot as baselines via a pipeline splitting inference, vision (Jetson Orin Nano), and motion control (Raspberry Pi 5).

Graduate Researcher, Applied Dynamics and Optimization Laboratory — Sept 2025 - July 2026
- Coupled COMSOL fluid simulations with MuJoCo rigid-body dynamics through an OSQP optimization program to build a gait cost analysis pipeline for a bio-inspired fish robot, matching CFD-derived loads within a relative error under 5%.
- Conducted a grid search over 9 parameters through the pipeline to inform gait parameter selection for the fish robot.
- Generated a bipedal walk cycle for the Unitree G1, prototyping inverse kinematics that solves center-of-mass, posture, and foot-frame objectives via quadratic programming, using Mink, producing a full walking gait for the humanoid.

Mechanical Engineer, Robotic Design Team — Jan 2025 - May 2025
- Designed and manufactured the excavation subsystem's frame and bucket for NYU's Lunabotics rover, using SolidWorks/Onshape for CAD and prototyping via CNC machining and 3D printing.
- Integrated the subsystem with the frame, coordinating as part of a 50-person cross-functional team, enabling it to pass preliminary qualification testing.

Senior Developer and Asset Designer, Developers' Society — May 2020 - Sep 2024
- Designed and integrated game-ready 3D assets for club productions in Blender and Unity, including a SAE car with a custom parametric shader and a campus digital twin.
- Reconstructed real-world objects and environments into high-fidelity game assets, using Gaussian splatting and NeRFs for 3D scene representation.
- Implemented ray tracing, ray marching, and volumetric rendering shaders using OpenGL for custom game rendering engines.

Undergraduate Researcher, Dynamics and Vibrations Lab Aug 2021 - Jan 2024

- Designed an adaptive suspension controller that tunes to varying road profiles, using an online-learning MLP trained via a leaky-average gradient-descent law with a replay buffer.
- Benchmarked classical control strategies such as PD, PID, LQR, sliding mode, and sky-hook in a custom MATLAB/Simulink framework, beating the strongest baseline by 17.4% overshoot and 63% peak force.

Research Scholar, Computational Intelligence Lab Jan 2023 - Aug 2023

- Estimated muscle activation from motion without invasive sensors, via a signal processing pipeline comparing EMG to OpenSim-simulated activation, matching measured EMG within 1-5% error.
- Validated the measurements across an OptiTrack/force-plate dataset spanning 5 subjects and 12 poses, producing a 2M-frame dataset.

PROJECTS

Quadruped RL Policy for Locomotion Sept 2025 - Present

- Trained a robust locomotion RL policy for the Unitree Go2 quadruped via PPO training, using a custom reward model implementing Raibert-heuristic, feet-clearance, and contact-force terms, on rough and stepped terrains.
- Transferred the trained policy onto the physical robot for sim-to-real deployment, using domain randomization and a PD actuator model to close the sim-to-real gap, achieving stable walking during in-lab testing.

Vision-Guided Manipulation & Gesture Teleoperation July 2026

- Engineered hands-free robotic arm control from a webcam, building a real-time MediaPipe hand pose estimator for an xArm 6, mapping hand motions to the end-effector via a rate-limited control loop, validated across 6 sessions.
- Automated cube pick-and-place and stacking with a vision-guided manipulation pipeline on the arm, using YOLO object detection and point cloud segmentation for localization and retrieval.

Vision-Based Maze Navigation Agent Sept 2025 - Dec 2025

- Solved a maze-navigation task using only first-person observation images, building a k-nearest-neighbor visual-similarity graph from the ResNet-50 embeddings of the observations to enable localization and navigation.
- Trained a traversal policy over the kNN graph through a tabular Q-learning agent with reward shaped by distance to target, succeeding in over 50% of navigation runs.
- Built a manual-evaluation pipeline for navigation using COLMAP, back-projecting depth maps into colored 3D point clouds.

Robot Perception and Computer Vision Sept 2025 - Dec 2025

- Implemented a pipeline to find previously seen locations for visual place recognition, building a content-based image retrieval system using ResNet-18 embeddings indexed in a KD-tree, achieving over 80% accuracy on the NYU-VPR dataset.
- Developed a multi-vehicle tracking script, implementing Lucas-Kanade-style optical flow, then clustering tracked keypoints with DBSCAN to produce per-vehicle motion trails.
- Built a point-cloud registration pipeline for 3D mapping, implementing ICP using nearest-neighbor correspondence and a closed-form rigid-transform solution, incorporating RANSAC for plane fitting, validated on KITTI LiDAR frames.

Optimal Control & Reinforcement Learning Jan 2024 - Dec 2025

- Automated exploration and goal-finding across a decentralized drone swarm, combining Reynolds flocking rules and potential-field obstacle avoidance for traversal with particle swarm optimization to propagate goal information across the group.
- Designed a constrained aerobatic maneuver for a quadrotor, tracking it with a receding-horizon MPC controller, validated over 25 10-second flights under disturbances.
- Controlled a 3-DOF manipulator model, deploying a Model Predictive Path Integral controller using Monte Carlo sampling over a black-box simulator, reaching target positions with around 1% run-to-run variability.

Gesture-Controlled Hexapod Robot Sept 2024 - Dec 2024

- Designed and built a gesture-controlled hexapod robot, writing Arduino firmware for gait/action behaviors like tripod walking, turning, and sit-stand, and interfacing an I2C/SPI motor driver alongside a custom Li-ion power circuit.
- Developed a wrist-mounted gesture controller using flex sensors and an ESP32, encoding and wirelessly transmitting the gestures to the hexapod.
- Integrated onboard object recognition using a Pi Camera interfaced to a Raspberry Pi 4.